Chapter 120

Transform Nodes

This chapter details the Transform nodes available in Fusion.

The abbreviations next to each node name can be used in the Select Tool dialog when searching for tools and in scripting references.

For purposes of this document, node trees showing MediaIn nodes in DaVinci Resolve are interchangeable with Loader nodes in Fusion Studio, unless otherwise noted.

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Camera Shake [CSh]

The Camera Shake node

Camera Shake Node Introduction

This node can simulate a variety of camera shake-style motions from organic to mechanical. It is not the same as the Shake Modifier, which generates random number values for parameters.

For more information on the Shake modifier, *see Chapter 124, "Modifiers,"* in the DaVinci Resolve Reference Manual, or Chapter 64 in the Fusion Reference Manual.

The Camera Shake node concatenates its result with adjacent transformation nodes for higher-quality processing.

Inputs

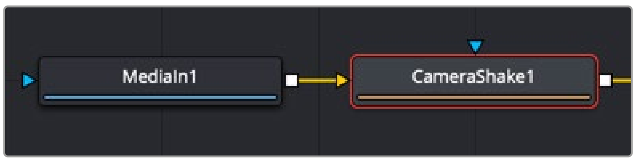
The two inputs on the Camera Shake node are used to connect a 2D image and an effect mask, which can be used to limit the camera shake area.

Input: The orange input is used for the primary 2D image that shakes.

Effect Mask: The blue input is for a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps from other tools. Connecting a mask to this input limits the camera shake area to only those pixels within the mask. An effects mask is applied to the tool after the tool is processed.

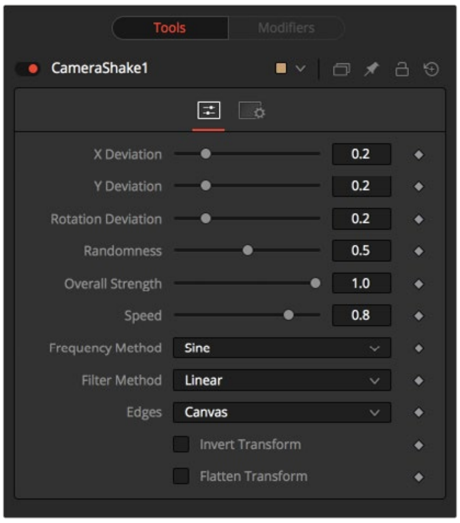
Basic Node Setup

The Camera Shake background input is used to connect the image you want to transform. Polygon masks can be connected into the occlusion mask input to identify areas the camera shake should ignore.



The Camera Shake node can be connected directly after a MediaIn node or any node providing a 2D output.

Inspector



The Camera Shake Controls tab

Controls Tab

The Controls tab includes parameters for adjusting the offsets, strength, speed, and frequency of the simulated camera shake movement.

Deviation X and Y

These controls determine the amount of shake applied to the image along the horizontal (X) and vertical (Y) axes. Values between 0.0 and 1.0 are permitted. A value of 1.0 generates shake positions anywhere within the boundaries of the image.

Rotation Deviation

This determines the amount of shake that is applied to the rotational axis. Values between 0.0 and 1.0 are permitted.

Randomness

Higher values in this control cause the movement of the shake to be more irregular or random. Smaller values cause the movement to be more predictable.

Overall Strength

This adjusts the general amplitude of all the parameters and blends that affect in and out. A value of 1.0 applies the effect as described by the remainder of the controls.

Speed

Speed controls the frequency, or rate, of the shake.

Frequency Method

This selects the overall shape of the shake. Available frequencies are Sine, Rectified Sine, and Square Wave. A Square Wave generates a much more mechanical-looking motion than a Sine.

Filter Method

When rescaling a pixel, surrounding pixels are often used to give a more realistic result. There are various algorithms for combining these pixels, called filters. More complex filters can give better results but are usually slower to calculate.

The best filter for the job often depends on the amount of scaling and on the contents of the image itself.

- **Box:** This is a simple interpolation resize of the image.
- **Linear:** This uses a simplistic filter, which produces relatively clean and fast results.
- **Quadratic:** This filter produces a nominal result. It offers a good compromise between speed and quality.
- **Cubic:** This produces better results with continuous-tone images. If the images have fine detail in them, the results may be blurrier than desired.
- **Catmull-Rom:** This produces good results with continuous-tone images that are resized down. This produces sharp results with finely detailed images.
- **Gaussian:** This is very similar in speed and quality to Bi-Cubic.
- **Mitchell:** This is similar to Catmull-Rom but produces better results with finely detailed images. It is slower than Catmull-Rom.
- **Lanczos:** This is very similar to Mitchell and Catmull-Rom but is a little cleaner and also slower.
- **Sinc:** This is an advanced filter that produces very sharp, detailed results; however, it may produce visible "ringing" in some situations.
- **Bessel:** This is similar to the Sinc filter but may be slightly faster.

Window Method (Sinc and Bessel Only)

Some filters, such as Sinc and Bessel, require an infinite number of pixels to calculate exactly. To speed up this operation, a windowing function is used to approximate the filter and limit the number of pixels required. This control appears when a filter that requires windowing is selected.

- **Hanning:** This is a simple tapered window.
- **Hamming:** Hamming is a slightly tweaked version of Hanning that does not taper all the way down to zero.
- **Blackman:** A window with a more sharply tapered falloff.
- **Kaiser:** A more complex window with results between Hamming and Blackman.

Most of these filters are useful only when making an image larger. When shrinking images, it is common to use the Bi-Linear filter, however, the Catmull-Rom filter will apply some sharpening to the results and may be useful for preserving detail when scaling down an image.

Example

Resize filters. From left to right: Nearest Neighbor, Box, Linear, Quadratic, Cubic, Catmull-Rom, Gaussian, Mitchell, Lanczos, Sinc, and Bessel.

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Edges

This menu determines how the Edges of the image are treated.

- **Canvas:** This causes the edges that are revealed by the shake to be the canvas color—usually transparent or black.
- **Wrap:** This causes the edges to wrap around (the top is wrapped to the bottom, the left is wrapped to the right, and so on).
- **Duplicate:** This causes the Edges to be duplicated, causing a slight smearing effect at the edges.
- **Mirror:** Image pixels are mirrored to fill to the edge of the frame.

Invert Transform

Select this control to Invert any position, rotation, or scaling transformation. This option might be useful for exactly removing the motion produced in an upstream Camera Shake.

Flatten Transform

The Flatten Transform option prevents this node from concatenating its transformation with adjacent nodes. The node may still concatenate transforms from its input, but it will not concatenate its transformation with the node at its output.

Common Controls

Settings Tab

The Settings tab in the Inspector is also duplicated in other Transform nodes. These common controls are described in detail at the end of this chapter in “The Common Controls” section.

Crop [CRP]

The Crop node

Crop Node Introduction

The Crop node can be used to cut out a portion of an image or to offset the image into a larger image area. However, unlike using a mask, this node actually changes the resolution of the image.

> **TIP:** You can crop an image in the viewer by activating the Allow Box Selection in the upper-left corner of the viewer while the Crop node is selected and viewed. Then, drag a crop rectangle around the area of interest to perform the operation.

> **NOTE:** Because this node changes the physical resolution of the image, animating the parameters is not advised.

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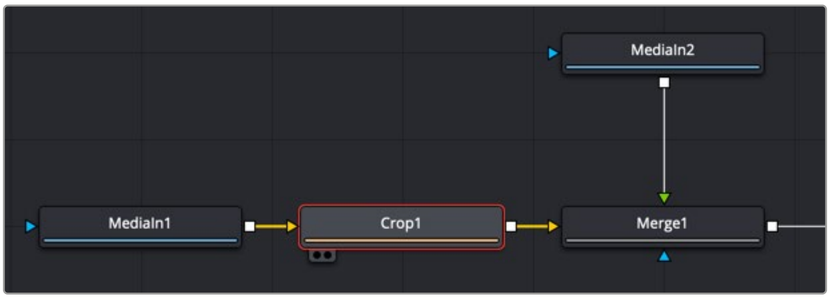
Inputs

The single input on the Crop node is used to connect a 2D image for cropping.

Input: The orange input is used for the primary 2D image you want to crop.

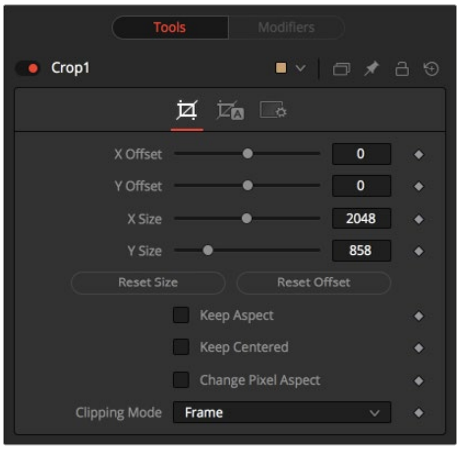
Basic Node Setup

Below, the Crop node is inserted between the MediaIn1 node and the background input of the Merge. Unlike using a mask tool, cropping the MediaIn1 changes the resolution of the clip. The cropped MediaIn1 node connected to the orange background input also sets the resolution of the Merge output.



The Crop node can be used to cut out a portion of an image.

Inspector



The Crop Controls tab

Controls Tab

The Controls tab provides XY Offset and XY Size methods for cropping the image.

Offset X and Y

These controls position the image off the screen by pushing it left/right or up/down. The cropped image disappears off the edges of the output image. The values of these controls are measured in pixels.

Size X and Y

Use these controls to set the vertical and horizontal resolution of the image output by the Crop node. The values of these controls are measured in pixels.

Keep Aspect

When toggled on, the Crop node maintains the aspect of the input image.

Keep Centered

When toggled on, the Crop node automatically adjusts the X and Y Offset controls to keep the image centered. The XY Offset sliders are automatically adjusted, and control over the cropping is done with the Size sliders or the Allow Box Selection button in the viewer.

Reset Size

This resets the image dimensions to the size of the input image.

Reset Offset

This resets the X and Y Offsets to their defaults.

Change Pixel Aspect

Enable this checkbox to reveal a Pixel Aspect control that can be used to change the image's pixel aspect.

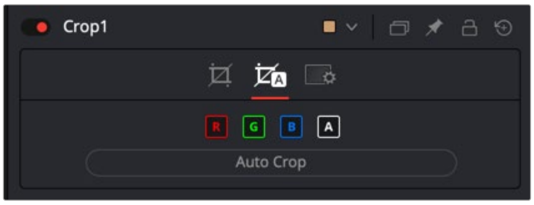
Clipping Mode

This option sets the mode used to handle the edges of the image when performing domain of definition (DoD) rendering. This is profoundly important for nodes like Blur, which may require samples from portions of the image outside the current domain.

- **Frame:** The default option is Frame, which automatically sets the node's domain of definition to use the full frame of the image, effectively ignoring the current domain of definition. If the upstream DoD is smaller than the frame, the remaining area in the frame will be treated as black/transparent.
- **Domain:** Setting this option to Domain will respect the upstream DoD when applying the node's effect. This can have adverse clipping effects in situations where the node employs a large filter.
- **None:** Setting this option to None does not perform any source image clipping at all. This means that any data required to process the node's effect that would normally be outside the upstream DoD is treated as black/transparent.

Auto Crop Tab

Auto Crop tab analyzes the selected channel and crops the image based on that channel's boundaries. The adjustments from auto crop are seen in the Crop tab parameters.



The Auto Crop tab

RGBA Color Channels

Select which channels are examined for an Auto Crop. This is useful for auto cropping images with non-solid backgrounds in a specific color channel, like a blue color gradient. Toggling the channel off causes Auto Crop to ignore it when evaluating the image.

Auto Crop

This evaluates the image and attempts to determine the background color. It then crops each side of the image to the first pixel that is not that color.

Common Controls

Settings Tab

The Settings tab in the Inspector is also duplicated in other Transform nodes. These common controls are described in detail at the end of this chapter in "The Common Controls" section.

DVE [DVE]

The DVE node

DVE Node Introduction

The DVE (Digital Video Effects) node is a 3D-image transformation similar to nodes found in old, tape-based online editing suites. The node encompasses image rotations, perspective changes, and Z moves. The axis can be defined for all transformations.

Inputs

The three inputs on the DVE node are used to connect a 2D image, DVE mask, and an effect mask, which can be used to limit the DVE area.

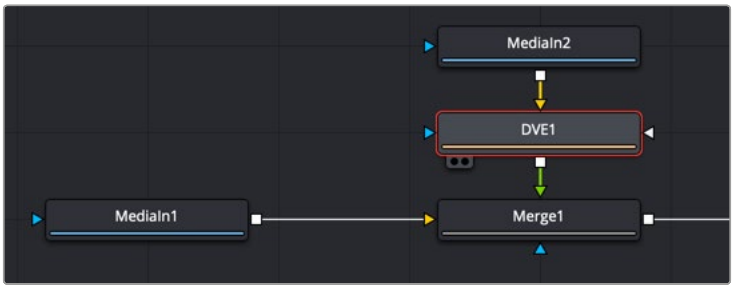
Input: The orange input is used for the primary 2D image that is transformed by the DVE.

DVE Mask: The white DVE mask input is used to mask the image prior to the DVE transform being applied. This has the effect of modifying both the image and the mask.

Effect Mask: The blue input is for a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps from other tools. Connecting a mask to this input causes the DVE to modify only the image within the mask. An effects mask is applied to the tool after the tool is processed.

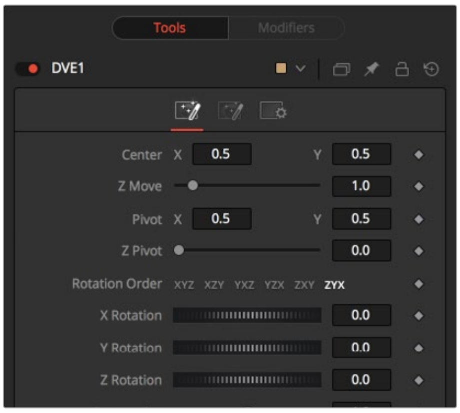
Basic Node Setup

In the example below, the DVE node is inserted between the MediaIn2 node and the foreground input of the Merge. The MediaIn1 node is manipulated in the DVE node and composited over the top of the MediaIn1 node.



The DVE node modifying the foreground input of a Merge node

Inspector



The DVE Controls tab

Controls Tab

The Controls tab includes all the transform parameters for the DVE.

Pivot X, Y, and Z

Positions the axis of rotation and scaling. The default is 0.5, 0.5 for X and Y, which is in the center of the image, and 0 for Z, which is at the center of Z space.

Rotation Order

Use these buttons to determine in what order rotations are applied to the image.

XYZ Rotation

These controls are used to rotate the image around the pivot along the X-, Y- and Z-axis.

Center X and Y

This positions the center of the DVE image onscreen. The default is 0.5, 0.5, which positions the DVE in the center of the image.

Z Move

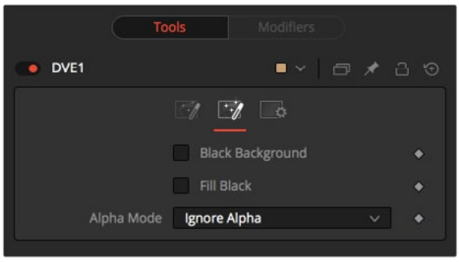
This zooms the image in and out along the Z-axis. Visually, when this control is animated, the effect is similar to watching an object approach from a distance.

Perspective

This adds additional perspective to an image rotated along the X- or Y-axis, similar to changing the Field of View and zoom of a camera.

Masking Tab

The DVE node allows pre-masking of its input image. This offers the ability to create transformations from the masked area of the image while leaving the remainder of the image unaffected.



The DVE Masking tab

Unlike regular effect masks, the masking process occurs before the transformation. All the usual mask types can be applied to the DVE mask.

Black Background

Toggle this on to erase the area outside the mask from the transformed image.

Fill Black

Toggle this on to erase the area within the mask (before transformation) from the DVE's input, effectively cutting the masked area out of the image. Enabling both Black Background and Fill Black will show only the masked, transformed area.

Alpha Mode

This determines how the DVE will handle the alpha channel of the image when merging the transformed image areas over the untransformed image.

- **Ignore Alpha:** This causes the input image's alpha channel to be ignored, so all masked areas will be opaque.
- **Subtractive/Additive:** These cause the internal merge of the pre-masked DVE image over the input image to be either Subtractive or Additive.

  An Additive setting is necessary when the foreground DVE image is premultiplied, meaning that the pixels in the color channels have been multiplied by the pixels in the alpha channel. The result is that transparent pixels are always black, since any number multiplied by 0 always equals 0. This obscures the background (by multiplying with the inverse of the foreground alpha), and then simply adds the pixels from the foreground.

  A Subtractive setting is necessary if the foreground DVE image is not premultiplied. The compositing method is similar to an Additive merge, but the foreground DVE image is first multiplied by its own alpha, to eliminate any background pixels outside the alpha area.

Common Controls

Settings Tab

The Settings tab in the Inspector is also duplicated in other Transform nodes. These common controls are described in detail at the end of this chapter in "The Common Controls" section.

Letterbox [LBX]

The Letterbox node

Letterbox Node Introduction

Use the Letterbox node to adapt existing images to the frame size and aspect ratios of any other format. The most common use of this node is to convert film resolution images to HD-sized frames for viewing on an external television monitor. Horizontal or vertical black edges are automatically added where necessary to compensate for aspect ratio differences. This node actually changes the resolution of the image.

NOTE: Because this node changes the physical resolution of the image, animating the controls is not recommended.

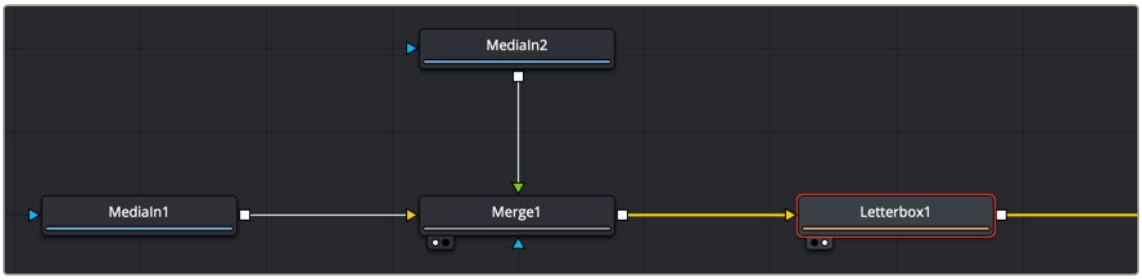
Inputs

The single input on the Letterbox node is used to connect a 2D image for letterbox/cropping.

Input: The orange input is used for the primary 2D image you want to letterbox/crop.

Basic Node Setup

The Letterbox node is used in the example below to change the resolution of the Merge node's output. Depending on how the resolution is modified, side pillars, or a horizontal letterbox mask, is applied to "fill in" the frame area, which the Merge node output does not cover.



The Letterbox node converting the Merge output resolution and adding letterbox masking where needed.

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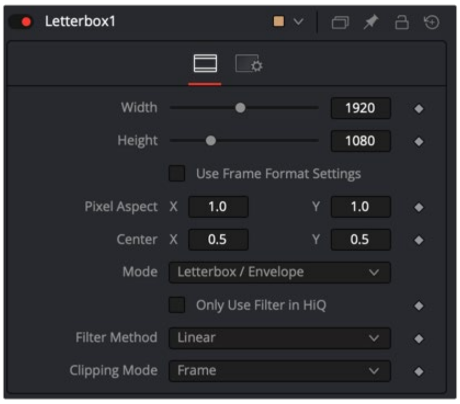
Inspector



The Letterbox Controls tab

Controls Tab

The Controls tab includes parameters for adjusting the resolution and pixel aspect of the image. It also has the option of letterboxing or pan-and-scan formatting.

Width and Height

The values of these controls determine the size of the output image as measured in pixels.

TIP: You can use the formatting contextual menu to quickly select a resolution from a list. Place the pointer over the Width or Height controls, and then right-click to display the contextual menu. The bottom of the menu displays a Select Frame Format submenu with available frame formats. Select any one of the choices from the menu to set the Height, Width, and Aspect controls automatically.

Auto Resolution

Activating this checkbox automatically sets the Width and Height sliders to the Frame Format settings found in the Preferences window for Fusion Studio or to the resolution of the DaVinci Resolve Timeline.

Pixel Aspect X and Y

These controls determine the pixel aspect ratio of the output image.

Center X and Y

This Center control repositions the image window when used in conjunction with Pan-and-Scan mode. It has no effect on the image when the node is set to Letterbox mode.

Mode

This control is used to determine the Letterbox node's mode of operation.

- **Letterbox/Envelope:** This corrects the aspect of the input image and resizes it to match the specified width.
- **Pan-and-Scan:** This corrects the aspect of the input image and resizes it to match the specified height. If the resized input image is wider than the specified width, the Center control can be used to animate the visible portion of the resized input.

Filter Method

When rescaling a pixel, surrounding pixels are often used to give a more realistic result. There are various algorithms for combining these pixels, called filters. More complex filters can give better results but are usually slower to calculate. The best filter for the job often depends on the amount of scaling and on the contents of the image itself.

- **Box:** This is a simple interpolation resize of the image.
- **Linear:** This uses a simplistic filter, which produces relatively clean and fast results.
- **Quadratic:** This filter produces a nominal result. It offers a good compromise between speed and quality.
- **Cubic:** This produces better results with continuous-tone images. If the images have fine detail in them, the results may be blurrier than desired.
- **Catmull-Rom:** This produces good results with continuous-tone images that are resized down. This produces sharp results with finely detailed images.
- **Gaussian:** This is very similar in speed and quality to Bi-Cubic.
- **Mitchell:** This is similar to Catmull-Rom but produces better results with finely detailed images. It is slower than Catmull-Rom.
- **Lanczos:** This is very similar to Mitchell and Catmull-Rom but is a little cleaner and also slower.
- **Sinc:** This is an advanced filter that produces very sharp, detailed results; however, it may produce visible “ringing” in some situations.
- **Bessel:** This is similar to the Sinc filter but may be slightly faster.

Window Method (Sinc and Bessel Only)

Some filters, such as Sinc and Bessel, require an infinite number of pixels to calculate exactly. To speed up this operation, a windowing function is used to approximate the filter and limit the number of pixels required. This control appears when a filter that requires windowing is selected.

- **Hanning:** This is a simple tapered window.
- **Hamming:** Hamming is a slightly tweaked version of Hanning that does not taper all the way down to zero.
- **Blackman:** A window with a more sharply tapered falloff.
- **Kaiser:** A more complex window with results between Hamming and Blackman.

Most of these filters are useful only when making an image larger. When shrinking images, it is common to use the Bi-Linear filter; however, the Catmull-Rom filter will apply some sharpening to the results and may be useful for preserving detail when scaling down an image.

Example

Different resize filters. From left to right: Nearest Neighbor, Box, Linear, Quadratic, Cubic, Catmull-Rom, Gaussian, Mitchell, Lanczos, Sinc, and Bessel.

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Common Controls

Settings Tab

The Settings tab in the Inspector is also duplicated in other Transform nodes. These common controls are described in detail at the end of this chapter in “The Common Controls” section.

Planar Transform [Pxf]

The Planar Transform node

The Planar Transform node applies perspective distortions generated by a Planar Tracker node onto any input mask or masked image. The Planar Transform node can be used to reduce the amount of time spent on rotoscoping objects. The workflow here centers around the notion that the Planar Tracker node can be used to track objects that are only roughly planar. After an object is tracked, a Planar Transform node can then be used to warp a rotospline, making it approximately follow the object over time. Fine-level cleanup work on the rotospline then must be done.

Depending on how well the Planar Tracker followed the object, this can result in substantial time savings in the amount of tedious rotoscoping. The key to using this technique is recognizing situations where the Planar Tracker performs well on an object that needs to be rotoscoped.

A rough outline of the workflow involved is:

1. Track: Using a Planar Tracker node, select a pattern that represents the object to be rotoscoped. Track the shot (see the tracking workflow in the Track section for the Planar Tracker node).
2. Create a Planar Transform node: Press the Create Planar Transform button on the Planar Tracker node to do this. The newly created Planar Transform node can be freely cut and pasted into another composition as desired.
3. Rotoscope the object: Move to any frame that was tracked by the Planar Tracker. When unsure if a frame was tracked, look in the Spline Editor for a tracking keyframe on the Planar Transform node. Connect a Polygon node into the Planar Transform node. While viewing the Planar Transform node, rotoscope the object.
4. Refine: Scrub the timeline to see how well the polygon follows the object. Adjust the polyline on frames where it is off. It is possible to add new points to further refine the polygon.

Inputs

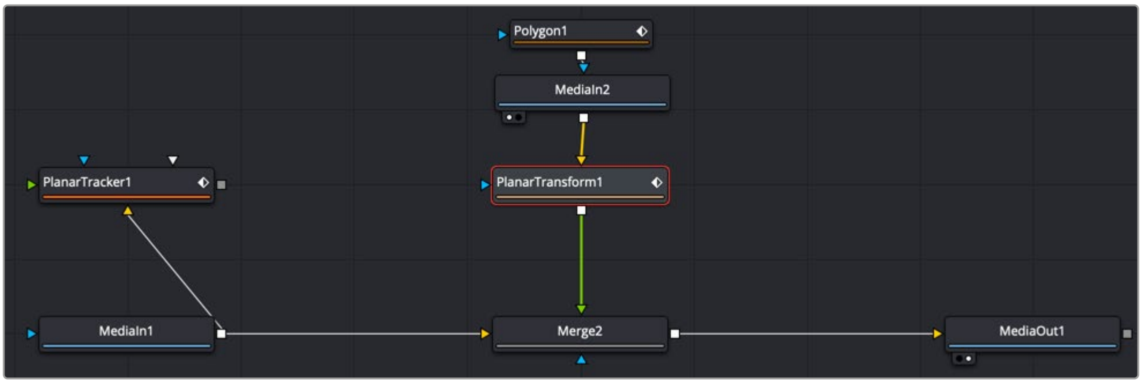
The Planar Transform has only two inputs:

Image Input: The orange image input accepts a 2D image on which the transform will be applied.

Effect Mask: The blue input is for a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps from other tools. Connecting a mask to this input limits the output of the Planar Transform to certain areas.

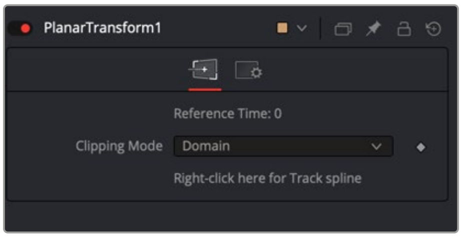
Basic Node Setup

The example below uses a Planar Transform node between a masked MediaIn2 node and the foreground input to a Merge node. The background MediaIn1 node connects to the Planar Tracker, which was used to generate the Planar Transform. Once the Planar Transform is created, the Planar Tracker is no longer needed in the node tree.



A Planar Transform creating a match move

Inspector



The Planar Transform Controls tab

Controls Tab

The Planar Transform node has very few controls, and they are all located in the Controls tab. It's designed to apply the analyzed Planar Tracking data as a match move,

Reference Time

This is the reference time that the pattern was taken from in the Planar Tracker node used to produce the Planar Transform.

Right-Click Here for Track Spline

The Track spline contains information about the perspective distortions stored in 4 x 4 matrices. When a Planar Transform node is exported from a Planar Tracker node, the Track spline produced by the Planar Tracker is shared by connecting it with the Planar Transform node. A consequence of this sharing of the Track spline is that if the track is changed in the Planar Tracker node, the Planar Transform will be automatically updated. Note that this spline can be examined in the Spline Editor, which is useful for seeing the extent of tracked frames.

Common Controls

Settings Tab

The Settings tab in the Inspector is also duplicated in other Tracking nodes. These common controls are described in detail at the end of this chapter in "The Common Controls" section.

Resize [RSZ]

The Resize node

Resize Node Introduction

Use the Resize node to increase or decrease the resolution of an input image. This is useful for converting images from one format to another (for example, from film to video resolution). This node actually changes the resolution of the image.

NOTE: Because this node changes the physical resolution of the image, animating the controls is not advised.

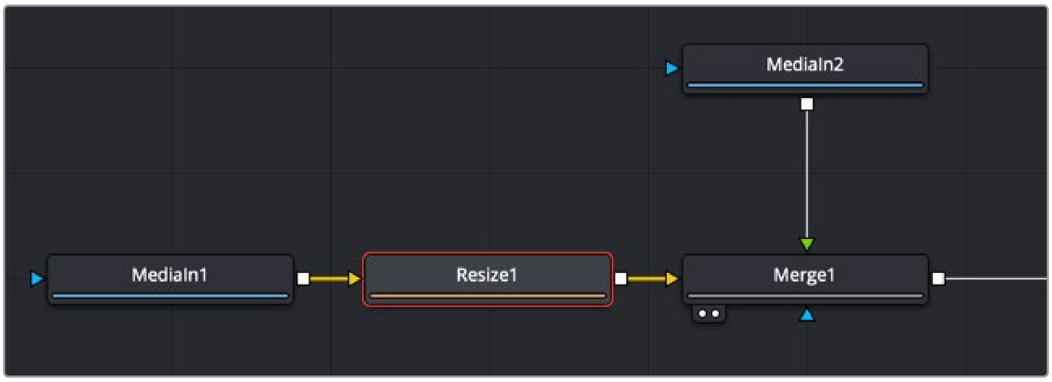
Inputs

The single input on the Resize node is used to connect a 2D image for resizing.

Input: The orange input is used for the primary 2D image you want to resize.

Basic Node Setup

Below, the Resize node is inserted between the MediaIn1 node and the background input of the Merge. Unlike using a Transform tool, resizing the MediaIn1 changes the resolution of the clip. The resized MediaIn1 node connected to the orange background input also sets the resolution of the Merge output.



The Resize node can be used to scale an image and change its resolution.

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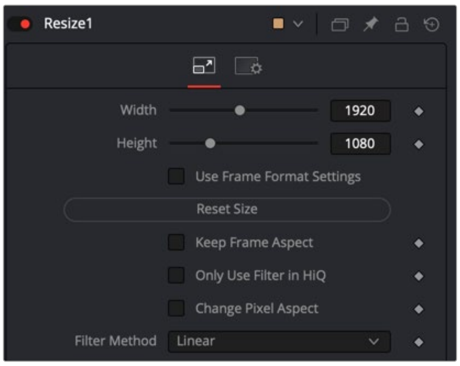
Inspector



The Resize Controls tab

Controls Tab

The Controls tab includes parameters for changing the resolution of the image. It uses pixel values in the Width and Height controls.

Width

This controls the new resolution for the image along the X-axis.

Height

This controls the new resolution for the image along the Y-axis.

> **TIP:** You can use the formatting contextual menu to quickly select a resolution from a list. Place the mouse pointer over the Width or Height controls, and then right-click to display the contextual menu. The bottom of the menu displays a Select Frame Format submenu with available frame formats. Select any one of the choices from the menu to set the Height and Width controls automatically.

Auto Resolution

Activating this checkbox automatically sets the Width and Height sliders to the Frame Format settings found in the Preferences window for Fusion Studio or the resolution in the DaVinci Resolve Timeline.

Reset Size

Resets the image dimensions to the original size of the image.

Keep Frame Aspect

When toggled on, the Resize node maintains the aspect of the original image, preserving the original ratio between width and height.

Only Use Filter in HiQ

The Resize node will normally use the fast Nearest Neighbor filter for any non-HiQ renders, where speed is more important than full accuracy. Disable this checkbox to force Resize to always use the selected filter for all renders.

Change Pixel Aspect

Enable this checkbox to reveal a Pixel Aspect control that can be used to change the image's pixel aspect.

Filter Method

When rescaling a pixel, surrounding pixels are often used to give a more realistic result. There are various algorithms for combining these pixels, called filters. More complex filters can give better results but are usually slower to calculate. The best filter for the job often depends on the amount of scaling and on the contents of the image itself.

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Window Method (Sinc and Bessel Only)

Some filters, such as Sinc and Bessel, require an infinite number of pixels to calculate exactly. To speed up this operation, a windowing function is used to approximate the filter and limit the number of pixels required. This control appears when a filter that requires windowing is selected.

- **Hanning:** This is a simple tapered window.
- **Hamming:** Hamming is a slightly tweaked version of Hanning that does not taper all the way down to zero.
- **Blackman:** A window with a more sharply tapered falloff.
- **Kaiser:** A more complex window with results between Hamming and Blackman.

Most of these filters are useful only when making an image larger. When shrinking images, it is common to use the Bi-Linear filter; however, the Catmull-Rom filter will apply some sharpening to the results and may be useful for preserving detail when scaling down an image.

Example

Different resize filters. From left to right: Nearest Neighbor, Box, Linear, Quadratic, Cubic, Catmull-Rom, Gaussian, Mitchell, Lanczos, Sinc, and Bessel.

Common Controls

Settings Tab

The Settings tab in the Inspector is also duplicated in other Transform nodes. These common controls are described in detail at the end of this chapter in “The Common Controls” section.

Scale [SCL]

The Scale node

Scale Node Introduction

The Scale node is almost identical to the Resize node, except that Resize uses exact dimensions, whereas the Scale node uses relative dimensions to describe the change to the source image's resolution. This node actually changes the resolution of the image.

NOTE: Because this node changes the physical resolution of the image, animating the controls is not advised.

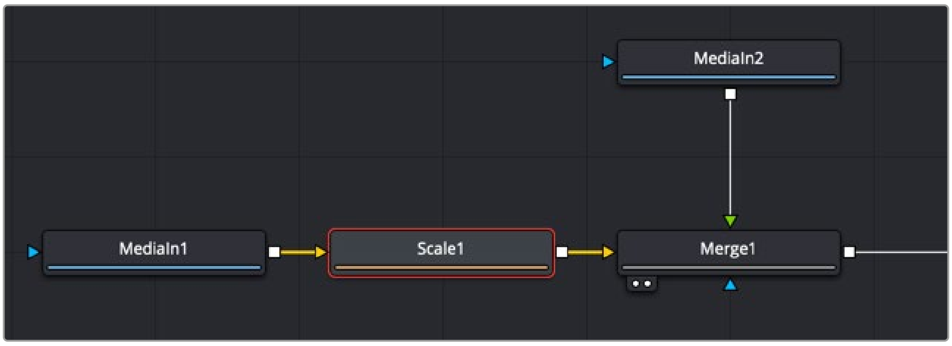
Inputs

The single input on the Scale node is used to connect a 2D image for scaling.

Input: The orange input is used for the primary 2D image you want to scale.

Basic Node Setup

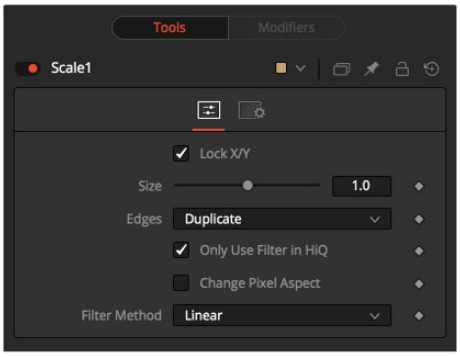
Below, the Scale node is inserted between the MediaIn1 node and the background input of the Merge. Unlike using a Transform tool, scaling the MediaIn1 changes the resolution of the clip. The resized MediaIn1 node connected to the orange background input also sets the resolution of the Merge output.



The Scale node can be used to scale an image and change its resolution.

FUSION

Inspector



The Scale Controls tab

Controls Tab

The Controls tab includes parameters for changing the resolution of the image. It uses a multiplier of size to set the new resolution. An Edges menu allows you to determine how the edges of the frame are handled if the scaling decreases.

Lock X/Y

When selected, only a Size control is shown, and changes to the image's scale are applied to both axes equally. If the checkbox is cleared, individual Size controls appear for both X and Y Size.

Size

The Size control is used to set the scale used to adjust the resolution of the source image. A value of 1.0 would have no affect on the image, while 2.0 would scale the image to twice its current resolution. A value of 0.5 would halve the image's resolution.

Only Use Filter in HiQ

The Scale node will normally use the fast Nearest Neighbor filter for any non-HiQ renders, where speed is more important than full accuracy. Disable this checkbox to force Scale to always use the selected filter for all renders.

Change Pixel Aspect

Enable this checkbox to reveal a Pixel Aspect control that can be used to change the image's pixel aspect.

Filter Method

When rescaling a pixel, surrounding pixels are often used to give a more realistic result. There are various algorithms for combining these pixels, called filters. More complex filters can give better results but are usually slower to calculate. The best filter for the job often depends on the amount of scaling and on the contents of the image itself.

- **Box:** This is a simple interpolation resize of the image.
- **Linear:** This uses a simplistic filter, which produces relatively clean and fast results.
- **Quadratic:** This filter produces a nominal result. It offers a good compromise between speed and quality.

- **Cubic:** This produces better results with continuous-tone images. If the images have fine detail in them, the results may be blurrier than desired.
- **Catmull-Rom:** This produces good results with continuous-tone images that are resized down. This produces sharp results with finely detailed images.
- **Gaussian:** This is very similar in speed and quality to Bi-Cubic.
- **Mitchell:** This is similar to Catmull-Rom but produces better results with finely detailed images. It is slower than Catmull-Rom.
- **Lanczos:** This is very similar to Mitchell and Catmull-Rom but is a little cleaner and also slower.
- **Sinc:** This is an advanced filter that produces very sharp, detailed results; however, it may produce visible "ringing" in some situations.
- **Bessel:** This is similar to the Sinc filter but may be slightly faster.

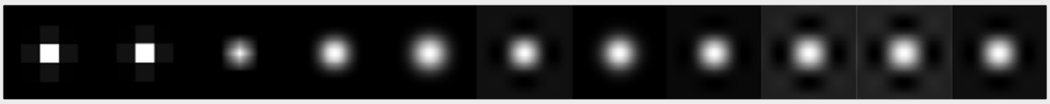
Window Method (Sinc and Bessel Only)

Some filters, such as Sinc and Bessel, require an infinite number of pixels to calculate exactly. To speed up this operation, a windowing function is used to approximate the filter and limit the number of pixels required. This control appears when a filter that requires windowing is selected.

- **Hanning:** This is a simple tapered window.
- **Hamming:** Hamming is a slightly tweaked version of Hanning that does not taper all the way down to zero.
- **Blackman:** A window with a more sharply tapered falloff.
- **Kaiser:** A more complex window with results between Hamming and Blackman.

Most of these filters are useful only when making an image larger. When shrinking images, it is common to use the Bi-Linear filter; however, the Catmull-Rom filter will apply some sharpening to the results and may be useful for preserving detail when scaling down an image.

Example



Different resize filters. From left to right: Nearest Neighbor, Box, Linear, Quadratic, Cubic, Catmull-Rom, Gaussian, Mitchell, Lanczos, Sinc, and Bessel.

NOTE: Because this node changes the physical resolution of the image, animating the controls is not advised.

Transform [XF]

The Transform node

Transform Node Introduction

The Transform node can be used for simple 2D transformations of the image, such as moving, rotating, and scaling. The image's aspect can also be modified using the Transform node.

The Transform node concatenates its result with adjacent Transformation nodes. The Transform node does not change the image's resolution.

Inputs

The two inputs on the Transform node are used to connect a 2D image and an effect mask, which can be used to limit the transformed area.

Input: The orange input is used for the primary 2D image that gets transformed.

Effect Mask: The blue input is for a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps from other tools. Connecting a mask to this input limits the transform area to only those pixels within the mask. An effects mask is applied to the tool after the tool is processed.

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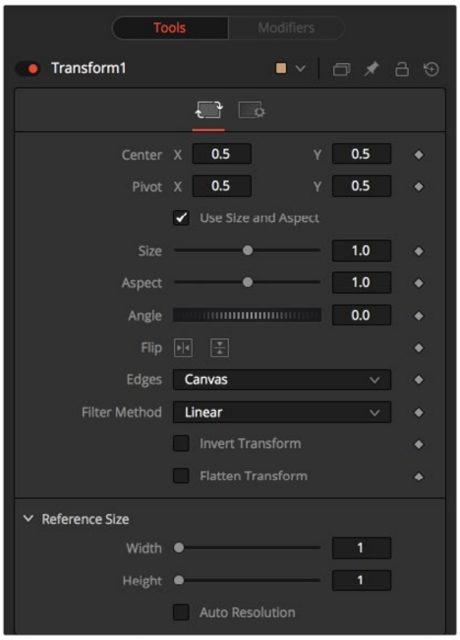
Basic Node Setup

The Transform node in the example below is inserted between the MediaIn2 node and the foreground input of the Merge. Unlike using a Scale or Resize tool, transforming the MediaIn2 does not change the resolution of the clip. For that reason, it is the tool most often used to scale, move, and rotate a clip.

The Transform node can be used to scale an image without changing its resolution.

Inspector



The Transform Controls tab

Controls Tab

The Controls tab presents multiple ways to transform, flip (vertical), flop (horizontal), scale, and rotate an image. It also includes reference size controls that can reinterpret the coordinates used for width and height from relative values of 0-1 into pixel values based on the image's resolution.

Center X and Y

This sets the position of the image on the screen. The default is 0.5, 0.5, which places the image in the center of the screen. The value shown is always the actual position multiplied by the reference size. See below for a description of the reference size.

Pivot X and Y

This positions the axis of rotation and scaling. The default is 0.5, 0.5, which is the center of the image.

Use Size and Aspect

This checkbox determines whether the Transform node provides independent Size controls for the X and Y scale or if Size and Aspect controls are used instead.

Size

This modifies the scale of the image. Values range from 0 to 5, but any value greater than zero can be entered into the edit box. If the Use Size and Aspect checkbox is selected, this control will scale the image equally along both axes. If the Use Size and Aspect option is off, independent control is provided for X and Y.

Aspect

This control changes the aspect ratio of an image. Setting the value above 1.0 stretches the image along the X-axis. Values between 0.0 and 1.0 stretch the image along the Y-axis. This control is available only when the Use Size and Aspect checkbox is enabled.

Angle

This control rotates the image around the axis. Increasing the Angle rotates the image in a counterclockwise direction. Decreasing the Angle rotates the image in a clockwise direction.

Flip Horizontally and Vertically

Toggle this control on to flip the image along the X- or Y-axis.

Edges

This menu determines how the edges of the image are treated when the edge of the raster is exposed.

- **Canvas:** This causes the edges of the image that are revealed to show the current Canvas Color. This defaults to black with no Alpha and can be set using the Set Canvas Color node.
- **Wrap:** This wraps the edges of the image around the borders of the image. This is useful for seamless images to be panned, creating an endless moving background image.
- **Duplicate:** This causes the edges of the image to be duplicated as best as possible, continuing the image beyond its original size.
- **Mirror:** Image pixels are mirrored to fill to the edge of the frame.

Filter Method

When rescaling a pixel, surrounding pixels are often used to give a more realistic result. There are various algorithms for combining these pixels, called filters. More complex filters can give better results but are usually slower to calculate. The best filter for the job often depends on the amount of scaling and on the contents of the image itself.

- **Box:** This is a simple interpolation resize of the image.
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Window Method (Sinc and Bessel Only)

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- **Hanning:** This is a simple tapered window.
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Most of these filters are useful only when making an image larger. When shrinking images, it is common to use the Bi-Linear filter; however, the Catmull-Rom filter will apply some sharpening to the results and may be useful for preserving detail when scaling down an image.

Example

Different resize filters. From left to right: Nearest Neighbor, Box, Linear, Quadratic, Cubic, Catmull-Rom, Gaussian, Mitchell, Lanczos, Sinc, and Bessel.

Invert Transform

Select this control to invert any position, rotation, or scaling transformation. This option is useful when connecting the Transform to the position of a tracker for the purpose of reintroducing motion back into a stabilized image.

Flatten Transform

The Flatten Transform option prevents this node from concatenating its transformation with adjacent nodes. The node may still concatenate transforms from its input, but it will not concatenate its transformation with the node at its output.

Reference Size

The controls under the Reference Size menu do not directly affect the image. Instead they allow you to control how Fusion represents the position of the Transform node's center.

Normally, coordinates are represented as values between 0 and 1, where 1 is a distance equal to the full width or height of the image. This allows for resolution independence, because you can change the size of the image without having to change the value of the center.

One disadvantage to this approach is that it complicates making pixel-accurate adjustments to an image. To demonstrate, imagine an image that is 100 x 100 pixels in size. To move the center of the image to the right by 5 pixels, we would change the X value of the transform center from 0.5, 0.5 to 0.55, 0.5. We know the change must be 0.05 because 5/100 = 0.05.

The Reference Size controls allow you to specify the dimensions of the image. This changes the way the control values are displayed, so that the Center shows the actual pixel positions in the X and Y number fields of the Center control. Extending our example, if you set the Width and Height to 100 each, the Center would now be shown as 50, 50, and we would move it 5 pixels toward the right by entering 55, 50.

Internally, the Transform node still stores this value as a number between 0 to 1, and if you were to query the Center controls value via scripting, or publish the Center control for use by other nodes, then you would retrieve the original normalized value. The change is visible only in the value shown for Transform Center in the node control.

Reference Width and Height Sliders

Set these to the width and height of the image to change the way that Fusion displays the values of the Transform node's Center control.

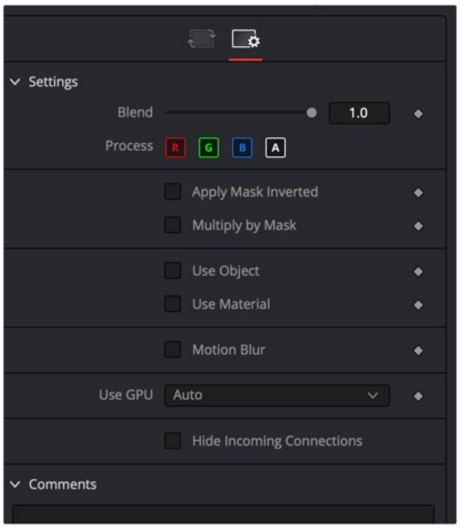
Auto Resolution

Enable this checkbox to use the current frame format settings in Fusion Studio or the timeline resolution in DaVinci Resolve to set the Reference Width and Reference Height values.

The Common Controls

Nodes that handle Transform operations share several identical controls in the Inspector. This section describes controls that are common among Transform nodes.

Inspector



The Transform Common controls

Settings Tab

The Settings tab in the Inspector can be found on every tool in the Transform category. The Settings controls are even found on third-party Transform-type plugin tools. The controls are consistent and work the same way for each tool.

Blend

The Blend control is used to blend between the tool's original image input and the tool's final modified output image. When the blend value is 0.0, the outgoing image is identical to the incoming image. Normally, this will cause the tool to skip processing entirely, copying the input straight to the output.

Process When Blend Is 0.0

The tool is processed even when the input value is zero. This can be useful if the node is scripted to trigger a task, but the node's value is set to 0.0.

Red/Green/Blue/Alpha Channel Selector

These four buttons are used to limit the effect of the tool to specified color channels. This filter is often applied after the tool has been processed.

For example, if the Red button on a Blur tool is deselected, the blur will first be applied to the image, and then the red channel from the original input will be copied back over the red channel of the result.

There are some exceptions, such as tools for which deselecting these channels causes the tool to skip processing that channel entirely. Tools that do this will generally possess a set of identical RGBA buttons on the Controls tab in the tool. In this case, the buttons in the Settings and the Controls tabs are identical.

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Apply Mask Inverted

Enabling the Apply Mask Inverted option inverts the complete mask channel for the tool. The mask channel is the combined result of all masks connected to or generated in a node.

Multiply by Mask

Selecting this option will cause the RGB values of the masked image to be multiplied by the mask channel's values. This will cause all pixels of the image not included in the mask (i.e., set to 0) to become black/transparent.

Use Object/Use Material (Checkboxes)

Some 3D software can render to file formats that support additional channels. Notably, the EXR file format supports Object ID and Material ID channels, which can be used as a mask for the effect. These checkboxes determine whether the channels will be used, if present. The specific Material ID or Object ID affected is chosen using the next set of controls.

Correct Edges

This checkbox appears only when the Use Object or Use Material checkboxes are selected. It toggles the method used to deal with overlapping edges of objects in a multi-object image. When enabled, the Coverage and Background Color channels are used to separate and improve the effect around the edge of the object. If this option is disabled (or no Coverage or Background Color channels are available), aliasing may occur on the edge of the mask.

> For more information on the Coverage and Background Color channels, *see Chapter 78, "Understanding Image Channels,"* in the DaVinci Resolve Reference Manual, or Chapter 18 in the Fusion Reference Manual.

Object ID/Material ID (Sliders)

Use these sliders to select which ID will be used to create a mask from the object or material channels of an image. Use the Sample button in the same way as the Color Picker: to grab IDs from the image displayed in the viewer. The image or sequence must have been rendered from a 3D software package with those channels included.

Motion Blur

- **Motion Blur:** This toggles the rendering of Motion Blur on the tool. When this control is toggled on, the tool's predicted motion is used to produce the motion blur caused by the virtual camera's shutter. When the control is toggled off, no motion blur is created.
- **Quality:** Quality determines the number of samples used to create the blur. A quality setting of 2 will cause Fusion to create two samples to either side of an object's actual motion. Larger values produce smoother results but increase the render time.
- **Shutter Angle:** Shutter Angle controls the angle of the virtual shutter used to produce the motion blur effect. Larger angles create more blur but increase the render times. A value of 360 is the equivalent of having the shutter open for one full frame exposure. Higher values are possible and can be used to create interesting effects.
- **Center Bias:** Center Bias modifies the position of the center of the motion blur. This allows for the creation of motion trail effects.
- **Sample Spread:** Adjusting this control modifies the weighting given to each sample. This affects the brightness of the samples.

Use GPU

The Use GPU menu has three settings. Setting the menu to Disable turns off GPU hardware-accelerated rendering. Enabled uses the GPU hardware for rendering the node. Auto uses a capable GPU if one is available and falls back to software rendering when a capable GPU is not available.

Hide Incoming Connections

Enabling this checkbox can hide connection lines from incoming nodes, making a node tree appear cleaner and easier to read. When enabled, empty fields for each input on a node will be displayed in the Inspector. Dragging a connected node from the node tree into the field will hide that incoming connection line as long as the node is not selected in the node tree. When the node is selected in the node tree, the line will reappear.

Comments

The Comments field is used to add notes to a tool. Click in the empty field and type the text. When a note is added to a tool, a small red square appears in the lower-left corner of the node when the full tile is displayed, or a small text bubble icon appears on the right when nodes are collapsed. To see the note in the Node Editor, hold the mouse pointer over the node to display the tooltip.

Scripts

Three Scripting fields are available on every tool in Fusion from the Settings tab. They each contain edit boxes used to add scripts that process when the tool is rendering. For more details on scripting nodes, please consult the Fusion scripting documentation.